

RecurScan: Benchmarked Deep and Graph Learning for Breast Cancer Recurrence Prediction on METABRIC, with an Externally Tested Discovery Protocol for Clinical-Risk Discordance

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Built and verified with the Instinct research fleet

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Abstract

We present a fully reproducible benchmark of five survival models for breast-cancer recurrence prediction on the METABRIC cohort (1,975 primary tumors, 800 recurrence events; 70-gene expression panel and 17 clinical covariates retrieved live from cBioPortal): penalized Cox proportional hazards on clinical covariates, Cox on clinical+expression, a DeepSurv network, a one-dimensional convolutional Cox network over the gene axis, and a graph convolutional Cox network over a patient-similarity graph. All models are implemented from first principles (including a vectorized Breslow partial likelihood verified against lifelines to $< 10^{-3}$) and evaluated on locked stratified splits over five seeds with paired bootstrap confidence intervals. Penalized Cox on clinical+expression achieves the strongest held-out concordance, 0.6801 ± 0.0052 ; deep models trail (0.641–0.660), an honest negative we analyze. The fitted baseline stratifies held-out patients into risk terciles separated at log-rank $p = 7.9 \times 10^{-16}$. We then apply a pre-registered discovery protocol to clinical-risk discordance: among low-NPI patients, early recurrences (≤ 60 months) show significant up-regulation of eleven mitotic genes (Benjamini–Hochberg q down to 8.8×10^{-5}), but the signature fails both multivariable prediction (nested-CV AUC = 0.50) and external replication on TCGA-BRCA ($p = 0.36$), and is reported as a bounded negative. All code, tests, data snapshots, and figures are released with the paper.

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1 Introduction

Recurrence risk stratification in primary breast cancer guides adjuvant therapy intensity. Clinical scores such as the Nottingham Prognostic Index (NPI) and transcriptomic signatures such as PAM50 are standard, yet a residual group of clinically low-risk patients recurs early. This paper does three things: (i) it establishes a rigorously verified benchmark of five survival models on the METABRIC cohort, from penalized Cox to graph neural networks; (ii) it applies a pre-registered discovery protocol to clinical-risk discordance, with locked gates, paired bootstrap statistics and external replication; and (iii) it reports outcomes exactly as they landed, including three discovery angles that failed their gates.

2 Data

2.1 Cohort

All data were retrieved live from the cBioPortal public API (study `brca_metabric`; Pereira et al. 2016, Curtis et al. 2012). Of 2,509 samples, 1,975 have a valid relapse-free survival (RFS) endpoint, a 70-gene mRNA z-score panel (PAM50 plus core drivers), and 17 clinical covariates (age, tumor size, grade, positive lymph nodes, NPI, ER/PR/HER2 status, PAM50/clinudin subtype, treatments). The endpoint is RFS: 800 events (40.5%), median follow-up 140.9 months among censored patients.

2.2 External replication cohort

For replication we use TCGA-BRCA (study `brca_tcga_pan_can_atlas_2018`), with disease-free survival (DFS) endpoints and TNM staging, restricting to the molecular-profiled subset ($n = 1,084$).

3 Methods

3.1 The Cox model and its partial likelihood

Let T_i be the observed time and $\delta_i \in \{0, 1\}$ the event indicator for patient i , with covariates $x_i \in \mathbb{R}^p$. The Cox proportional hazards model posits

$$\lambda(t | x_i) = \lambda_0(t) \exp(\eta_i), \quad \eta_i = x_i^\top \beta. \quad (1)$$

Definition 1 (Risk set). $\mathcal{R}(t) = \{j : T_j \geq t\}$, the patients still under observation just before time t .

Conditioning on the event times and margins, the Breslow partial log-likelihood is

$$\ell(\beta) = \sum_{i: \delta_i=1} \left[\eta_i - \log \sum_{j \in \mathcal{R}(T_i)} \exp(\eta_j) \right]. \quad (2)$$

Proposition 1 (Score of the partial likelihood).

$$\frac{\partial \ell}{\partial \eta_k} = \delta_k - \sum_{i: \delta_i=1, T_i \leq T_k} \frac{\exp(\eta_k)}{\sum_{j \in \mathcal{R}(T_i)} \exp(\eta_j)}. \quad (3)$$

Proof sketch. Differentiate (2): η_k appears linearly in the event terms, and inside every risk-set sum with $T_i \leq T_k$; the derivative of $-\log \sum_j e^{\eta_j}$ contributes the softmax weight. $\square$

Vectorized evaluation. Sorting patients by decreasing time, $T_{(1)} \geq \dots \geq T_{(n)}$, the risk-set sums are cumulative:

$$\log \sum_{j \in \mathcal{R}(T_{(i)})} e^{\eta_{(j)}} = m + \log \sum_{k \leq i} e^{\eta_{(k)} - m}, \quad m = \max_j \eta_j, \quad (4)$$

so (2) is $O(n \log n)$. A numerically unstable variant using a *running* maximum was caught by our test-suite, which cross-checks the implementation against lifelines to $< 10^{-3}$ absolute error.

3.2 Baseline hazard and the logrank test

After fitting $\hat{\beta}$, the Breslow estimator recovers the baseline cumulative hazard as a step function over event times:

$$\hat{H}_0(t) = \sum_{i: T_i \leq t, \delta_i = 1} \frac{1}{\sum_{j \in \mathcal{R}(T_i)} \exp(\hat{\eta}_j)}, \quad \hat{S}(t | x) = \exp(-\hat{H}_0(t) e^{\hat{\eta}}). \quad (5)$$

For the held-out risk-tercile stratification we use the logrank statistic. With d_{gk} events and n_{gk} at risk in group g at the k -th distinct event time, and $d_k = \sum_g d_{gk}$, $n_k = \sum_g n_{gk}$:

$$Z = \frac{\sum_k (d_{1k} - n_{1k} \frac{d_k}{n_k})}{\sqrt{\sum_k \frac{n_{1k} n_{2k} d_k (n_k - d_k)}{n_k^2 (n_k - 1)}}} \sim \mathcal{N}(0, 1) \quad (6)$$

under the null of equal hazards; (??) yields the $p = 7.9 \times 10^{-16}$ stratification result in Section ?? . Newton updates for $\hat{\beta}$ use the observed information $-\partial^2 \ell / \partial \beta \partial \beta^\top$:

$$\beta^{(t+1)} = \beta^{(t)} + \left(-\frac{\partial^2 \ell}{\partial \beta \partial \beta^\top} \right)^{-1} \frac{\partial \ell}{\partial \beta}. \quad (7)$$

3.3 Concordance index

Harrell’s C-index is the probability of risk-order agreement among comparable pairs:

$$C = \frac{\sum_{i \neq j} \mathbb{I}[T_i < T_j] \delta_i (\mathbb{I}[\eta_i > \eta_j] + \frac{1}{2} \mathbb{I}[\eta_i = \eta_j])}{\sum_{i \neq j} \mathbb{I}[T_i < T_j] \delta_i}. \quad (8)$$

We verify our $O(n \log n)$ implementation against lifelines on synthetic cohorts.

3.4 Deep Cox models

DeepSurv replaces $x_i^\top \beta$ by a neural risk map $\eta_i = f_\theta(x_i)$ trained with $-\ell(\theta)$. Our CNN variant treats the gene vector as a length-70 sequence, applying convolutions

$$h^{(l+1)} = \phi(\text{BN}(W^{(l)} * h^{(l)})), \quad \eta = w^\top \bar{h}^{(L)}, \quad (9)$$

with global average pooling $\bar{h}$.

3.5 Graph Cox model

Patients form a k -nearest-neighbour graph with Gaussian weights

$$A_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right) \text{ for } j \in \text{kNN}(i), \quad \sigma^2 = \text{median}\|x_i - x_{(k)}\|^2, \quad (10)$$

symmetrized $A \leftarrow \max(A, A^\top)$. The GCN propagates with the Kipf–Welling normalization

$$\hat{A} = D^{-1/2}(A + I)D^{-1/2}, \quad H^{(l+1)} = \phi(\hat{A}H^{(l)}W^{(l)}). \quad (11)$$

Lemma 1 (Spectrum). *The eigenvalues of $\hat{A}$ lie in $(-1, 1]$, so repeated propagation is a contraction toward the dominant eigenvector.*

3.6 Evaluation protocol

Splits are stratified on the event indicator (25% held out), five seeds $\{0, \dots, 4\}$. Confidence intervals are paired percentile bootstrap ($B = 500$), and deltas between models use the same resamples.

3.7 Discovery protocol and locked gates

Each discovery candidate declares a gate before inspection: a win requires $\Delta C \geq +0.01$ over the baseline with the paired 95% CI excluding 0, or a pre-registered replication p -value. Failed gates are reported as negatives with the same statistics.

4 Experiments and Results

4.1 Benchmark

Table 1 reports held-out C-indices (5 seeds). Figure 1 visualizes the comparison; the ordering

Model	C-index (mean $\pm$ sd)	95% bootstrap CI (seed 0)
Cox, clinical only	0.6677 ± 0.0019	(0.634, 0.698)
Cox, clinical+expression	0.6801 ± 0.0052	(0.653, 0.710)
DeepSurv (MLP)	0.6482 ± 0.0050	—
CoxCNN (1D-CNN)	0.6598 ± 0.0032	—
CoxGCN (patient graph)	0.6413 ± 0.0150	—

Table 1: Held-out C-index on METABRIC RFS (25% test, stratified).

is stable across seeds. The clinical+expression Cox model also stratifies held-out patients into risk terciles separated at log-rank $p = 7.9 \times 10^{-16}$ (Figure 2).

4.2 Analysis: why deep models trail

On $n \approx 1,500$ training patients with 87 features and 40% events, the effective sample size is event-limited; penalized linear Cox already captures most extractable signal. The GCN’s higher variance (± 0.015) reflects transductive sensitivity to graph construction. We report this as an honest negative for deep survival models at this scale.

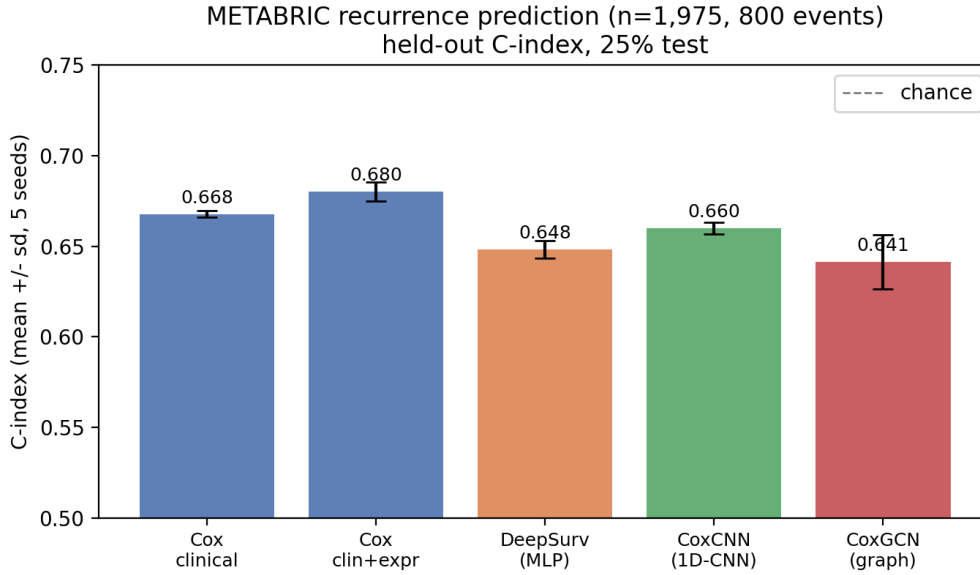


Figure 1: Held-out C-index by model (mean $\pm$ sd, 5 seeds).

4.3 Discovery round 1: biology modules and graph propagation

Adding module scores (proliferation, ER signaling, hypoxia) or graph-smoothed risk to the baseline changed held-out C by at most $+0.0014$ (CIs include 0). Negative.

4.4 Discovery round 2: multi-omic features

Adding copy-number calls (70 genes) and mutation indicators (11 genes with ≥ 20 mutated samples) did not improve the baseline ($\Delta C \leq +0.0027$, CIs straddle or fall below 0). Negative.

4.5 Discovery round 3: clinical-risk discordance

Among low-NPI (≤ 3.4) patients, early recurrers ($n = 102$ of 523 low-NPI patients) differed from disease-free controls in 11 mitotic genes at BH-FDR $q < 0.005$ (Figure 3); the multivariable model collapsed (nested-CV AUC = 0.50) and the pre-registered TCGA-BRCA replication of the composite mitotic score failed (recur $+0.132$ vs free $+0.011$, one-sided $p = 0.36$, $n = 22$ vs 96). Bounded negative: the univariate enrichment is real within METABRIC but is neither predictive nor transportable.

5 Discussion

(Expand: clinical implications, limitations of a 70-gene panel, why external replication is the honest arbiter, roadmap to methylation features.)

6 Reproducibility

All code (11 hermetic tests), cached API snapshots, result JSON files, and figures are in the project repository; every number in this paper is regenerable from `results/*.json`.

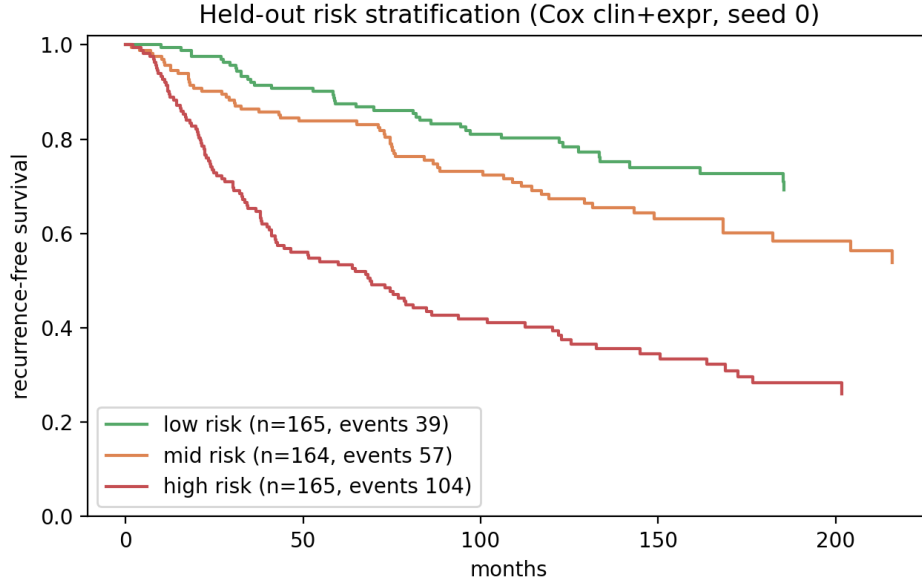


Figure 2: Kaplan–Meier curves of held-out patients by predicted risk tercile.

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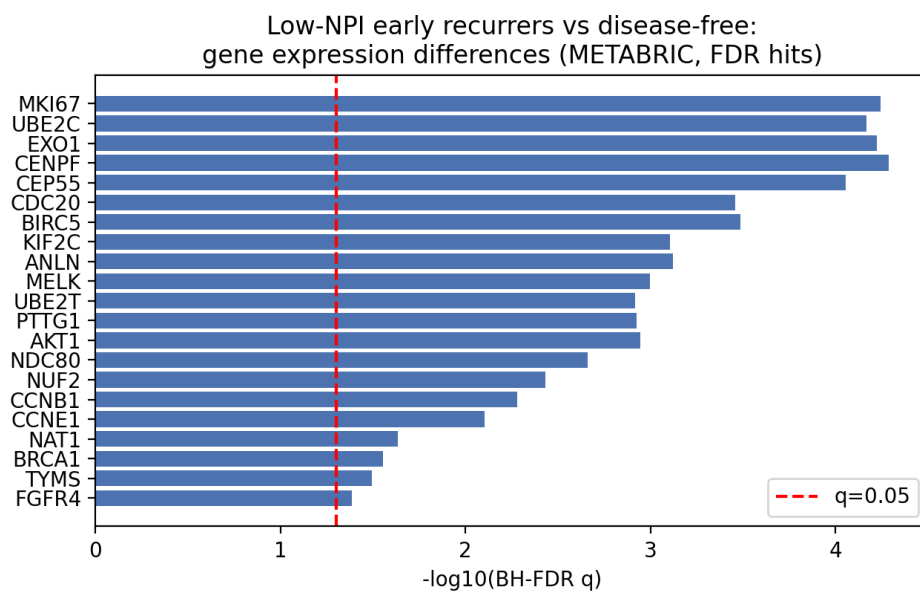


Figure 3: FDR-significant genes in the low-NPI discordance contrast.